



HOW THE SITE WORKS — End-to-End Traveler & Expedition Guide

Your Journey, Perfectly Mapped. Unified AI Itinerary Architecture, Live CARTO Cartography, 80+ Multi-Currency Engine with Country Flags, Circular Debt Simplification, Live Flight State Machines & Real-Time Collaboration.

Target Audience: Solo Nomads, Travel Groups, Expedition Leaders	Version: 1.0 Production-Ready
Stack: Next.js 16 (Turbopack), FastAPI, Leaflet, ReportLab	AI Engine: OpenRouter & Gemini Dual-Provider
Currencies Supported: 80+ Global Currencies with National Flags	Cartography: CARTO Dark Matter & Voyager, Satellite

1. Platform Overview & The 5-Stage Nomad Lifecycle

ARC-NOMADE replaces chaotic spreadsheets, disconnected chat threads, and currency confusion abroad with a single unified platform. The expedition lifecycle flows through five intuitive stages:

Stage	Phase Name	Key User Actions & Technologies
Phase 1	Account & Preferences	Set home airport, travel persona, budget tier, and preferred default currency from 80+ options with country flags.
Phase 2	Creation Wizard	4-step flow: Live geocoding pin-drop, date calculation, dynamic budget currency picker, and AI Travel Architect toggle.
Phase 3	Command Center	Unified trip dashboard with 5-day weather forecast, multi-tab agendas, CARTO map switcher, and live member presence.
Phase 4	Expedition Operations	Live flight state tracking, multi-currency conversion abroad, and automated circular debt simplification.
Phase 5	Dossier Archival	1-click exports: Executive 6-sheet Excel workbook with formulas and styled print-ready PDF travel dossiers.

2. Onboarding & Personalizing Your Travel Persona

Every traveler approaches expeditions differently. ARC-NOMADE tailors AI recommendations and default values based on your user profile (/profile):

- **Home Airport Code:** Configure standard 3-letter IATA codes (e.g., HND, JFK, LHR, CDG, SIN) to pre-populate flight routing.
- **Travel Personas:** Choose from *Balanced Explorer*, *Cultural Connoisseur*, *Foodie & Dining*, *Active & Outdoors*, or *Slow & Relaxed*.

- **Preferred Currency:** Select your domestic currency using the searchable CurrencySelect dropdown featuring 80+ currencies with national flags.
- **Budget Tier:** Tag your preference (\$ Budget Conscious, \$\$ Moderate Comfort, \$\$\$ High-End Luxury) to guide AI cost estimation.

3. The 4-Step Expedition Creation Wizard (/trips/create)

Creating an expedition takes under 60 seconds with real-time feedback at every step:

Step	Feature & Capability	What Happens Behind the Scenes
Step 1	Destination & Live Geocoding	Type any city or region (e.g. 'Kyoto, Japan'). Open-Meteo geocodes coordinates, and the interactive map preview automatically flies to the city and pins the exact location.
Step 2	Dates & Scenic Cover Photo	Pick start and end dates. Duration in days is computed automatically. Choose a curated scenic preset photo or provide a custom Unsplash cover URL.
Step 3	Budget, 80+ Currencies & AI Architect	Input Total Budget with dynamic currency prefix (\$ / € / £ / ¥ / ■ / etc.). Choose Primary Currency from 80+ countries with national flags. Toggle AI Travel Architect with desired daily pace.
Step 4	Review & Launch Expedition	Review summary card with spatial coordinates, duration, and country-flagged budget. Click 'Launch Expedition' for seamless navigation to the trip workspace.

4. Day-by-Day Itinerary Planner & Dual AI Architect

The Itinerary Planner structures your journey chronologically with rich multimedia cards:

- **Day-by-Day Agenda:** Grouped into expandable day cards with calendar date references and daily estimated cost subtotals.
- **Rich Item Details:** Each stop includes title, category (Sightseeing, Culinary, Transport, Accommodation, Activity, Relaxation), start/end time, physical address, and notes.
- **Dual AI Engine (OpenRouter & Gemini):** Click 'Generate with AI' to trigger automated itinerary synthesis. OpenRouter uses google/gemini-2.5-flash (with fallback to gpt-4o-mini) and a 4,500 token budget to output complete timetables with geocoded stop coordinates.

5. Interactive Spatial Cartography & CARTO Basemaps

The interactive Leaflet mapping engine visualizes every stop and route in real time:

- **Dual Pin Indicator:** Scheduled itinerary items render as illuminated **Cyan markers**; local AI-recommended discoveries render as **Purple markers**.
- **Connected Route Polylines:** Automatic route vectors link sequential stops throughout the day so travelers visualize travel order.
- **Live Basemap Switcher:** Switch between **CARTO Dark Matter** (sleek dark mode), **CARTO Voyager** (high-visibility explorer road map), **OpenStreetMap Standard**, and **Esri Satellite**.
- **Zero-Watermark Delivery:** Authenticated tile requests pass ?key=CARTO_KEY for crystal-clear cartographic display.
- **Performance Memoization:** Guarded by locationsRef and memoized position watchers to eliminate React re-render loops.

6. 80+ Multi-Currency Conversion Engine & Traveler Cheat Sheet

Traveling across borders introduces currency friction. ARC-NOMADE includes an integrated multi-currency conversion suite:

Component	Core Capabilities & User Flow
80+ Currencies & Flags	Full global coverage across Europe, Americas, Asia, Africa, Middle East, and Oceania. High-definition national flags loaded from flagcdn.com with emoji/code fallbacks.
CurrencySelect Dropdown	Searchable selector with instant fuzzy matching by code ('JPY'), country ('Japan'), or symbol ('¥'). Includes quick-access chips for USD, EUR, GBP, JPY, CAD, AUD, INR, CHF, SGD, AED, and THB.
 Traveler Cheat Sheet	Built-in quick reference matrix converting benchmark values (\$1, \$5, \$10, \$20, \$50, \$100, \$250, \$500, \$1,000) into destination currency for instant decision-making in taxis and cafes.
In-Modal Conversion	When adding an expense abroad, toggle the 'Paid in foreign currency?' drawer. Enter the amount paid in local currency; it converts automatically and logs the exchange rate in notes.

7. Expense Ledger & Circular Debt Simplification

Managing shared expenses in group travel usually leads to endless settlement friction. ARC-NOMADE automates fair splits with mathematical precision:

- **Split Flexibility:** Split costs Equally among selected travelers, assign Exact amounts, or allocate by Percentages.
- **Greedy Minimum Cash Flow:** Instead of Alice paying Bob who pays Charlie who pays Alice, the algorithm resolves net balances to the minimum possible number of peer-to-peer bank/cash transfers.
- **Settlement Matrix:** Visual dashboard showing exact net balances (green badge for creditors, red badge for debtors) and a 1-click 'Settle Payment' modal to record transactions.

8. Flight Operations & Live Boarding Passes

Keep flight itineraries organized with photorealistic digital boarding passes:

- **Flight Logging:** Record airline code, flight number, departure/arrival airports (with IATA codes), gate, terminal, and seat.
- **Live State Machine Simulator:** Automatic transition between operational flight states: *Scheduled* → *Boarding* → *Departed* → *In-Flight* → *Landed*.
- **Boarding Pass Cards:** Visual airline cards featuring barcode visuals, seat badges, terminal callouts, and countdown timers.

9. Real-Time Collaboration & Executive Dossier Exports

Expeditions require seamless team communication and offline document generation:

- **WebSocket Team Chat:** Real-time messaging with live avatar presence, typing indicators, and emoji reactions.
- **Role-Based Access Control (RBAC):** Grant OWNER, EDITOR, VIEWER, or EXPENSE_MANAGER permissions via shareable invite links.
- **Executive 6-Sheet Excel Dossier:** 1-click export of a formula-driven workbook featuring: Trip Overview, Itinerary Timetable, Flight Logistics, Expense Ledger (with live =SUM() formulas), Member Balances, and Category Breakdowns.
- **Print-Ready PDF Travel Dossier:** Generates a high-resolution, styled multi-page travel briefing ready for offline carry-on.

10. Quick Reference & Nomad Pro-Tips

Feature Area	Pro-Tip / Best Practice
Currency Selector	Press Escape to close the currency dropdown instantly. Use the popular chips row for 1-click selection of top travel currencies.
CARTO Basemaps	Toggle between CARTO Dark Matter for nighttime planning and CARTO Voyager or Satellite for outdoor trail exploration.
Expenses Abroad	Use the 'Paid in foreign currency?' drawer when buying coffee or train tickets abroad to let ARC-NOMADE auto-calculate the conversion.
Offline Readiness	Always export both the 6-Sheet Excel workbook and the PDF dossier before boarding long-haul flights with limited Wi-Fi.

Ready to explore the world? Launch your first journey at <http://localhost:3000/trips/create> and let ARC-NOMADE map your adventure with effortless precision.